

SUMS OF LARGE PRIME GAPS, THE GUTH–MAYNARD ESTIMATE, AND ANDRICA’S CONJECTURE

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ABSTRACT. Let $g_n = p_{n+1} - p_n$. Combining Järvinieni’s Harman-sieve refinement of Heath-Brown’s method with the large values estimate of Guth and Maynard, together with a few sharpenings of the sieve bookkeeping, we prove

$$\sum_{\substack{x \leq p_n \leq 2x \\ g_n \geq x^{1/2}}} g_n \ll x^{1/2+0.0575+\varepsilon},$$

improving the exponent $0.57 = 1/2 + 0.07$ of Järvinieni. Consequently the number of n with $p_n \leq X$ and $\sqrt{p_{n+1}} - \sqrt{p_n} \geq 1$ (failures of Andrica’s conjecture) is $\ll X^{0.0575+\varepsilon}$, and the number of $m \leq N$ for which $(m^2, (m+1)^2)$ contains no prime is $\ll N^{0.115+\varepsilon}$. The admissible ranges of Dirichlet polynomial lengths are established by a branch-and-bound certificate over a linear model, with a fixed power margin far above floating-point error, and the sieve loss is bounded by computer. We also construct a Beurling generalized prime system whose zeta function is zero-free with growth $|t|^{o(1)}$ on $\sigma > 1/2$ and whose integer and prime counting functions have error terms $O(x^{1/2+\varepsilon})$, yet which violates Andrica’s inequality infinitely often, so that no argument using only these properties can prove Andrica’s conjecture; the construction does not model zeros on the critical line. Finally we record a single-scale fourth-moment bound that would imply the conjecture.

1. INTRODUCTION

Write p_n for the n th prime and $g_n = p_{n+1} - p_n$. Heath-Brown [9] showed that

$$\sum_{x \leq p_n \leq 2x, g_n \geq x^{1/2}} g_n \ll x^{3/5+\varepsilon},$$

and Järvinieni [10] improved this to $x^{0.57+\varepsilon}$ by inserting Harman’s sieve into Heath-Brown’s argument. Järvinieni remarks [10, Remark 7.6] that non-rigorous sampling indicates his method cannot go much below 0.57, and that reaching $x^{0.56}$ “would require new ideas”. The new idea used here is the large values estimate of Guth and Maynard [6], which is strongest exactly when a Dirichlet polynomial of length N takes values near $N^{3/4}$, the critical configuration in the relevant range.

Theorem 1.1. *For every fixed $\varepsilon > 0$,*

$$\sum_{\substack{p_n \in [x, 2x] \\ g_n \geq x^{1/2}}} g_n \ll_{\varepsilon} x^{1/2+0.0575+\varepsilon}.$$

Corollary 1.2. *The number of n with $p_n \leq X$ and $\sqrt{p_{n+1}} - \sqrt{p_n} \geq 1$ is $\ll_{\varepsilon} X^{0.0575+\varepsilon}$.*

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Proof. $\sqrt{p_{n+1}} - \sqrt{p_n} \geq 1$ is equivalent to $g_n \geq 2\sqrt{p_n} + 1$. If $p_n \in [x, 2x]$ this forces $g_n \geq 2x^{1/2}$, so each such n contributes at least $2x^{1/2}$ to the sum in Theorem 1.1. Hence there are $\ll x^{0.0575+\varepsilon}$ such n with $p_n \in [x, 2x]$; summing over dyadic $x \leq X$ gives the claim. $\square$

Corollary 1.3. *The number of $m \leq N$ such that the interval $(m^2, (m+1)^2)$ contains no prime is $\ll_\varepsilon N^{0.115+\varepsilon}$.*

Proof. Fix a dyadic range $m^2 \in [x, 2x]$. If $(m^2, (m+1)^2)$ has no prime, it lies inside a gap (p_n, p_{n+1}) with $g_n \geq 2m+1 > x^{1/2}$; here $p_n < m^2 \leq 2x$, and $p_n > p_{n+1}/2 > x/2$ by Bertrand's postulate. A gap of length g contains at most $g/(2\sqrt{x}) + 1 \leq 2g/\sqrt{x}$ such intervals with $m^2 \geq x$. Theorem 1.1 (applied to the two dyadic ranges covering $[x/2, 2x]$) bounds the count by $\ll x^{-1/2} x^{1/2+0.0575+\varepsilon} = x^{0.0575+\varepsilon}$. Summing over dyadic $x \leq N^2$ gives the claim. $\square$

Gafni and Tao [5] feed the Guth–Maynard estimate into bounds for the number of exceptional intervals for the prime number theorem in short intervals; at length $x^{1/2}$ their unconditional results do not bound the sum in Theorem 1.1. The exponent database of Tao, Trudgian and Yang [1] lists 0.57 as the record for this sum. Corollary 1.2 is of course far from Andrica's conjecture itself, which remains open; the conjecture is known numerically for $p < 2^{64}$ [12]. With Järvinen's exponent the bound in Corollary 1.2 was $X^{0.07+\varepsilon}$, and with Heath-Brown's $X^{0.1+\varepsilon}$.

What is new. We follow [10] line by line and change only the following.

- (1) The parameter R is $x^{r+\nu}$ for a general $r \in [0.0575, 0.07]$; all lemmas of [10] that depend on r are checked for general r (Section 3). One region in [10, Lemma 7.4] needs an extra constraint when $r < 1/16$.
- (2) The admissible ranges for factorizations $F = ABC$ without zeta factors ([10, Proposition 6.1(i),(ii)]) are supplemented by a computer-certified admissible set, obtained from Huxley's large values estimate together with Theorem 2.1 (Section 4).
- (3) In the sieve computation, sums with two prime variables are tested directly against the admissible set, the factorization search runs over all groupings, and one prime polynomial at a time may be split by the Heath-Brown identity (Section 5). The first two steps use only the reduction of [10, Section 4], which applies to any number of prime variables and any grouping of the factors, and the third is permitted by [10, Remark 4.10]. A fourth step splits the remaining factor W into two groups.
- (4) Section 7 constructs a Beurling prime system whose zeta function is zero-free with growth $|t|^{o(1)}$ on $\sigma > 1/2$, with error terms $O(x^{1/2+\varepsilon})$ for N and ψ , that violates Andrica's inequality infinitely often, and proves that a fourth-moment bound for primes in intervals of length $\sqrt{X}$ implies Andrica's conjecture.

2. NOTATION AND THE INPUT FROM GUTH–MAYNARD

We keep the notation of [10]: $c = 1/2$, $T = x^{1/2}S^2$, $H' = x^{1/2}(\log \log x)^{-4}$, $L_\zeta = x^{1/4}$, $S = \exp((\log \log x)^{17})$, and $X \lesssim Y$ means $S^\varepsilon X \ll Y$ for some $\varepsilon > 0$. Now $R = x^{r+\nu}$. We work with logarithmic lengths: for a Dirichlet polynomial P of length $P = x^{p+o(1)}$ we write $p = \log_x P$.

The task ([10, Claim 5.1]) is: for every $F = ABC$ arising from the decomposition of [10, Information 4.15] and every $M(s) = \sum_{i \leq R} \zeta_i m_i^{-s}$ as in [10, Proposition 2.1],

$$(1) \quad \int_{T_1}^{2T_1} |F(it)M(it)| dt \lesssim Rx \quad ([T_1, 2T_1] \subset [T_0, T]).$$

Theorem 2.1 (Guth–Maynard [6, Theorem 1.1]). *Let $|b_n| \leq 1$ and let $(t_r)_{r \leq R_0}$ be 1-separated points of $[0, T]$ with $|\sum_{N \leq n \leq 2N} b_n n^{it_r}| \geq V$ for all r . Then*

$$R_0 \leq T^{o(1)} \left(N^2 V^{-2} + N^{18/5} V^{-4} + T N^{12/5} V^{-4} \right).$$

Lemma 2.2. *Let $P(s)$ be a product of factors of F , let $w \geq 1$ be a fixed integer, and let $\sigma \leq 1$. Let $\mathcal{T} \subset [T_1, 2T_1]$ be the set where $|P(it)| \sim P^\sigma$. Then for every fixed $\delta > 0$,*

$$|\mathcal{T}| \ll x^\delta \left(P^{w(2-2\sigma)} + P^{w(18/5-4\sigma)} + T P^{w(12/5-4\sigma)} \right).$$

Proof. Put $Q = P^w$. By [10, Lemma 5.3] and the fact that powers and products of prime polynomials have coefficients $O_w(1)$, the coefficients of Q are bounded by $K = O_w(1)S^{o(1)}$ when no factor is a zeta sum, and by $K = x^{o(1)}$ in general (since $|\xi_0(h)| \leq \tau(h)$), and Q is supported on $[Q, QS^{o(1)}]$. Split the support into $O(\log S)$ dyadic blocks. If $|Q(it)| \geq V := Q^\sigma/2$ then some block has absolute value $\geq V/O(\log S)$. Take a maximal 1-separated subset of $\mathcal{T}$; $|\mathcal{T}|$ is at most twice its cardinality. Applying Theorem 2.1 to each block with coefficients divided by K and with V replaced by $V/(K O(\log S))$ gives the bound with an extra factor $K^{O(1)}S^{o(1)}T^{o(1)} \leq x^\delta$, since $T_1 \leq T \leq x$. (For blocks of length $N \geq T$ the classical mean value theorem already gives $\ll N^2 V^{-2} \log x$, so the uniformity of the $o(1)$ in Theorem 2.1 is only needed for $N < T$.) $\square$

The loss x^δ in Lemma 2.2 is not of size $S^{o(1)}$, so every use below is made with a fixed power margin (Definition 4.2).

3. THE PARAMETER r IN JÄRVINIEMI'S LEMMAS

In [10] the numbers 0.43, 0.57, 0.56, 0.32 and 0.07 stand for $1/2 - r$, $1/2 + r$, $8r$, $1/4 + r$ and r with $r = 0.07$. We record where r enters.

Lemma 3.1. *For every $r \in (0, 0.07]$ the following statements of [10] hold with these substitutions: Proposition 6.1(i), (ii), (iii); Proposition 2.2 in the simplified form $R \leq T^{1/3}$ (valid as $r < 1/6$); Lemmas 5.8, 5.9, 7.1, 7.2, 7.3. Lemma 7.4 holds for the region*

$$B_1(r) = \{(\alpha, \beta) : \alpha > \beta > \tfrac{1}{4} + \epsilon, \alpha + \beta < \tfrac{1}{2} + r - \epsilon, \alpha + \beta > 1 - 8r + \epsilon, \\ \alpha + 0.7\beta > \tfrac{1}{2} - r + \epsilon, \alpha - 3\beta > -8r + \epsilon\}$$

and $B_2(r) = \{(\alpha, \beta) : (1 - \alpha - \beta, \beta) \in B_1(r)\}$. For $r \geq 1/16$ the added condition $\alpha + \beta > 1 - 8r$ is implied by $\alpha > \beta > 1/4$.

Proof. Proposition 6.1(i): $A, B \geq x^{1/2-r}$ gives $AC \leq xS^\epsilon/B \leq H'R$; the rest of the argument does not involve R . Proposition 6.1(ii): the first case needs $A^{2-2\sigma_A} \leq H'RA^{1-2\sigma_A}C^{1-2\sigma_C}$ with $\sigma_C \leq 4/5$, i.e. $AC^{3/5} \leq x^{1/2+r}$; the second case (with $w = 4$) needs $T^{3/4}A^{1/8}C \leq H'R$ and $TA^{1/8} \leq H'R$, i.e. $a \leq 1 + 8r - 8c$ and $a \leq 8r$, which is $A \leq x^{8r} \min(1, x/C^8)$; and $AC \geq H'R$ follows from $B < x^{1/2-r}$. Proposition 6.1(iii) uses $T^{1/2}C < H'R$ and $T < H'R$, i.e. $c < 1/4 + r$ and $r > 0$. Lemma 5.8 uses only $|M| \leq R$ and large values of the factors, as noted in [10]. In Lemma 5.9 the terms not involving R are unchanged, and the two error terms $x^{2\epsilon}\sqrt{R}T\sqrt{x}$ and $x^\epsilon T\sqrt{x}R^{3/4}$ are $\ll Rx/S^\epsilon$ for every fixed $r > 0$: since $T\sqrt{x} = xS^2$ and $R = x^{r+\nu}$, this amounts to $x^{2\epsilon}S^3 \leq R^{1/2}$ and $x^\epsilon S^3 \leq R^{1/4}$, which hold once $\epsilon < r/8$. In Lemma 7.1 the two constraints are: when $H > L_\zeta$, the impossibility of both $B_{i-1}R_i > x^{1/2}S^{-3}$ and $C_{i-1}R_i > x^{1/4+r}$ follows from $R_i \leq z < x^r$, since otherwise $HB_{i-1}C_{i-1}R_i > x^{1/4+1/2+1/4+r}/R_i > x$; when $M \geq x^{1/2-r}$ the last factor satisfies $XS^{-\epsilon}/(MR_k) > x^{1/2-r}$ because $M < x^{1/2}$ and $R_k < x^r$; when $M < x^{1/2-r}$ one has $MNHR_k < x^{(1/2-r)+(1/4+r)+1/4} = x$. Lemmas 7.2 and 7.3 use the same inequalities. In Lemma 7.4 the conditions used are $C = P < x^{1/4+r}$ (from $\alpha + \beta < 1/2 + r$,

$\beta > 1/4$), $PQ^{7/10} > x^{1/2-r}$, $x/(PQ) < x^{8r}$ and $PQ^{-3} > x^{-8r}$. In [10] the third was deduced from $PQ > x^{1/2} > x^{1-8r}$, which requires $r \geq 1/16$; for smaller r it is the added constraint $\alpha + \beta > 1 - 8r$. $\square$

4. ADMISSIBLE LENGTHS VIA AN EXACT CERTIFICATE

Assume $F = ABC$ has no zeta factor. Following the end of [10, Section 5], split $[T_1, 2T_1]$ into $S^{o(1)}$ sets $\mathcal{T}_\sigma$ on which $|A(it)| \sim A^{\sigma_A}$, $|B(it)| \sim B^{\sigma_B}$, $|C(it)| \sim C^{\sigma_C}$; the set where some factor is below x^{-1} is negligible, and $\sigma_A, \sigma_B, \sigma_C \leq 4/5$ by [10, Lemma 5.8]. Write a, b, c for the logarithmic lengths and $v_A = a\sigma_A$, etc., for the logarithmic sizes, and $\tau = \log_x |\mathcal{T}_\sigma|$. Then (up to $o(1)$) $a + b + c = 1$ and

$$(2) \quad -1 \leq v_Y \leq \frac{4}{5}y \quad (Y \in \{A, B, C\}).$$

Routes. By [10, (5.7), (5.8)], (1) holds on $\mathcal{T}_\sigma$ if $\tau \leq \rho - \delta$ for some fixed $\delta > 0$, where ρ is one of

$$\begin{aligned} \rho_0 &= \sum_Y (y - v_Y), \\ \rho_X &= \min \left(\sum_{Y \neq X} (2y - 2v_Y), \frac{1}{2} + r + \sum_{Y \neq X} (y - 2v_Y) \right) \quad (X \in \{A, B, C\}). \end{aligned}$$

The first is the L^1 bound; ρ_X comes from Cauchy–Schwarz and Heath-Brown’s mean value theorem [8], in the form [10, Proposition 2.2], applied to XM . More generally, for $Q = A^{k_A} B^{k_B} C^{k_C}$ with $0 \leq k_Y \leq 3$, the inequality $\int_{\mathcal{T}_\sigma} |FM| \leq \sup_{\mathcal{T}_\sigma} |F/Q| |\mathcal{T}_\sigma|^{1/2} (\int_0^T |QM|^2)^{1/2}$ and [10, Proposition 2.2] (which places no restriction on the length of Q) give the routes

$$\rho_Q = \min \left(2 - 2q - 2v_F + 2v_Q, \frac{3}{2} + r - q - 2v_F + 2v_Q \right), \quad q = \sum k_Y y, \quad v_Q = \sum k_Y v_Y, \quad v_F = \sum v_Y;$$

$k = e_X$ recovers ρ_X . The coefficients of Q are bounded by $x^{o(1)}$, which the fixed margin below absorbs.

Bounds for τ . For every $Y \in \{A, B, C\}$ and integer $1 \leq w \leq 30$, Huxley’s estimate ([10, Lemma 5.10 and (5.9)]) and Lemma 2.2 give, up to $x^{o(1)}$,

$$\begin{aligned} \tau &\leq \frac{1}{2}, \\ \tau &\leq \max \left(w(2y - 2v_Y), \frac{1}{2} + \min(w(y - 2v_Y), w(4y - 6v_Y)) \right), \\ \tau &\leq \max \left(w(2y - 2v_Y), w\left(\frac{18}{5}y - 4v_Y\right), \frac{1}{2} + w\left(\frac{12}{5}y - 4v_Y\right) \right). \end{aligned}$$

The form [10, (5.9)] of Huxley’s estimate needs $G \ll PS^{o(1)}$ for $P = Y^w$, where G is the sum of the squared moduli of the coefficients. For factors shorter than $x^{1/4}$ this is [10, Lemma 5.3(ii)]. A factor longer than $x^{1/4}$ without a zeta sum is a prime polynomial, and an integer has at most $w!$ ordered factorizations into w primes, so powers and products of prime polynomials and of factors shorter than $x^{1/4}$ have coefficients $O_w(1)S^{o(1)}$; hence $G \ll PS^{o(1)}$ in every case used here.

Remark 4.1. The same argument applies when one group contains a zeta sum: its coefficients are bounded by $\tau(h) = x^{o(1)}$, Huxley’s and the Guth–Maynard estimates apply to its powers after normalisation, and [10, Lemma 5.8] holds for any factorisation. All losses are $x^{o(1)}$ and are absorbed by the fixed margin δ .

Definition 4.2. Fix $\delta = 10^{-6}$. A point (a, c, v_A, v_B, v_C) is *certified* if some bound β from the list above and some route ρ satisfy $\beta \leq \rho - \delta$ there. A closed cell $Q = [a_0, a_1] \times [c_0, c_1]$ is *admissible* if every point with $(a, c) \in Q$, $b = 1 - a - c \geq 0$ and (v_A, v_B, v_C) satisfying (2) is certified.

Proposition 4.3. *Let $F = ABC$ have no zeta factor, each of A, B, C being either constant or of length at least z_1 , and suppose $(\log_x A, \log_x C)$ lies, up to $o(1)$, in an admissible cell. Then (1) holds.*

Proof. If one group is constant, say $c = v_C = 0$, the formulas remain valid with that group contributing nothing. For each of the $S^{o(1)}$ sets $\mathcal{T}_\sigma$ the point (a, c, v_A, v_B, v_C) is certified, with margin δ . The losses are $S^{o(1)}$ from Huxley's estimate and $x^{\delta/2}$ from Lemma 2.2, and the $o(1)$ perturbation of (a, b, c) moves the true point by $o(1)$; after clamping each v_Y into $[-1, \frac{4}{5}y]$ for the cell point (another $o(1)$ move), each linear form changes by $o(1)$. Hence $|\mathcal{T}_\sigma| \leq x^{\rho-\delta/3}$, which is (1) on $\mathcal{T}_\sigma$ by [10, (5.7)–(5.8)]. $\square$

Deciding admissibility. Every bound and every route is a maximum or minimum of affine functions of (a, c, v_A, v_B, v_C) . On an axis-parallel box the exact range of an affine function is computed by interval arithmetic with no overestimation. For a box $\mathcal{B}$, a pair (β, ρ) with $\beta = \max_i \min_j \ell_{ij}$ and $\rho = \min_k m_k$ certifies all of $\mathcal{B}$ if for every i and k there is j with $\sup_{\mathcal{B}}(\ell_{ij} - m_k) \leq -\delta$. Our program subdivides a cell (with v_Y ranging over $[-1, \frac{4}{5}y_{\max}]$) by bisection along the coordinate of largest weighted width until every sub-box is certified, lies outside the domain ($b < 0$ or some $v_Y > \frac{4}{5}y$ throughout), or a node budget is exhausted. A cell is marked admissible only in the first case; exhausting the budget, or finding an uncertified point, marks it not admissible. Before bisection, 9×9^3 sample points per cell are tested and any failure marks the cell not admissible. The grid has 400×400 cells of side $1/400$ in (a, c) .

The certificate uses only the sup of affine functions over boxes, so floating point errors are of order 10^{-15} , far below the margin δ .

A second grid for one zeta factor. Now let $F = ZBC$, where Z is a zeta sum of length x^z with $z > 1/4$ and B, C have no zeta factor. Write $\tau_1 = \log_x T_1 \leq 1/2 + o(1)$. If the coefficients of Z are ξ_0 or $\xi_0 \log$ and $Z \geq T_1 z_2$, then (1) holds by [10, Lemma 5.4], so we may assume $Z < T_1 z_2$, that is $z \leq \tau_1 + o(1)$. If the coefficients are $\xi_0 g$, then Z is supported on proper prime powers, so $|Z(it)| \leq Z^{1/2+o(1)}$ and $\int_{T_1}^{2T_1} |Z(it)|^{2k} dt \leq x^{1/2+o(1)} Z^k$, and each bound on Z listed below holds trivially. For ξ_0 and $\xi_0 \log$ the inputs are the following.

- (Z1) $v_Z \leq \frac{4}{5}z$ by [10, Lemma 5.8], and $v_Z \leq z/2 + 1/12$. For the latter, expand ξ_0 as in the proof of [10, Lemma 5.4] and apply to each sum $\sum_{n \sim N} n^{-it}$ with $N \leq Z$ the second-derivative bound $T_1^{1/2}$ and the third-derivative bound $N^{1/2} T_1^{1/6} + N T_1^{-1/6}$. With $z \leq \tau_1 \leq 1/2$, the third-derivative bound gives $z/2 + \tau_1/6 \leq z/2 + 1/12$ when $z \leq 2\tau_1/3$, and otherwise the second-derivative bound gives $\tau_1/2 \leq z/2 + 1/12$, because $\tau_1 \leq z + 1/6$ follows from $z > 2\tau_1/3$ and $\tau_1 \leq 1/2$.
- (Z2) $\int_{T_1}^{2T_1} |Z(it)|^{2k} dt \leq x^{o(1)} J_{2k}$, where $J_2 = x^{1/2} Z$ comes from the mean value theorem, which gives $\ll (T_1 + Z) Z x^{o(1)}$, together with $Z \leq T_1 z_2$; $J_4 = x^{1/2} Z^2$ is [10, Lemma 5.7]; $J_{12} = x Z^6$ follows from Heath-Brown's twelfth moment $\int_0^T |\zeta(\frac{1}{2} + it)|^{12} dt \ll T^2 \log^{17} T$ [7], transferred to Z by Perron's formula and Hölder's inequality as in [10, Lemmas 5.5 and 5.6]; and $J_6 = J_4^{3/4} J_{12}^{1/4}$, $J_8 = J_4^{1/2} J_{12}^{1/2}$ by Hölder's inequality. In logarithmic form $j_2 = \frac{1}{2} + z$, $j_4 = \frac{1}{2} + 2z$, $j_6 = \frac{5}{8} + 3z$, $j_8 = \frac{3}{4} + 4z$ and $j_{12} = 1 + 6z$.

The coefficients of Z and of its powers are bounded by $x^{o(1)}$, which is enough for Huxley's estimate, Lemma 2.2 and [10, Proposition 2.2], because every use below carries a fixed power margin.

With variables (z, c, v_Z, v_B, v_C) , $b = 1 - z - c$ and $\tau = \log_x |\mathcal{T}_\sigma|$, the *zeta domain* is given by $1/4 \leq z \leq 1/2$, $b \geq 0$, $-1 \leq v_Y \leq \frac{4}{5}y$ for $Y \in \{Z, B, C\}$, and $v_Z \leq z/2 + 1/12$. The *zeta bounds* for τ are $\tau \leq \frac{1}{2}$, the Huxley and Guth–Maynard bounds above for Y^w with $Y \in \{Z, B, C\}$ and

$1 \leq w \leq 30$, and the moment bounds

$$\tau \leq \frac{1}{2} + 2z - 4v_Z, \quad \tau \leq 1 + 6z - 12v_Z,$$

which follow from (Z2) since $|Z(it)| \geq x^{v_Z+o(1)}$ on $\mathcal{T}_\sigma$.

The *zeta routes* come from Hölder's inequality. A route is specified by a set $\mathcal{X} \subsetneq \{Z, B, C\}$ and a mode $\mu \in \{0, 2, 4\}$ for M , with $\mu = 0$ only when $\mathcal{X} = \emptyset$; by a mode $k_Y \in \{0, 1, 2, 3\}$ for each $Y \in \{B, C\} \setminus \mathcal{X}$; and by a mode $e \in \{0, 2, 4, 6, 8, 12\}$ for Z when $Z \notin \mathcal{X}$. Mode 0 means that the factor is bounded pointwise on $\mathcal{T}_\sigma$; mode $\mu = 2$ or 4 means that XM with $X = \prod_{Y \in \mathcal{X}} Y$ is taken in $L^\mu(0, T)$; $k_Y \geq 1$ means that Y^{k_Y} is taken in $L^2(T_1, 2T_1)$; and $e \geq 2$ means that Z is taken in $L^e(T_1, 2T_1)$. The Hölder weights are $1/\mu$, $1/(2k_Y)$ and $1/e$ for the non-pointwise modes, their sum W must satisfy $W \leq 1$, and $\alpha = 1 - W$ is the exponent of $|\mathcal{T}_\sigma|$. With $x_\mathcal{X} = \sum_{Y \in \mathcal{X}} y$, the factors contribute, in $\log_x$,

$$\begin{aligned} M : \quad & r \quad (\mu = 0), \quad \max(x_\mathcal{X} + r, \tfrac{1}{2}(x_\mathcal{X} + r + \tfrac{1}{2})) \quad (\mu = 2), \\ & \max(x_\mathcal{X} + r, \tfrac{1}{2}x_\mathcal{X} + \tfrac{3}{4}r + \tfrac{1}{8}) \quad (\mu = 4); \\ Y \in \{B, C\} \setminus \mathcal{X} : \quad & v_Y \quad (k_Y = 0), \quad \max(y, \tfrac{1}{2}y + \tfrac{1}{4k_Y}) \quad (k_Y \geq 1); \\ Z \notin \mathcal{X} : \quad & v_Z \quad (e = 0), \quad j_e/e \quad (e \geq 2). \end{aligned}$$

The M terms come from [10, Proposition 2.2], which gives $\int_0^T |XM|^2 \ll x^{o(1)}(X^2 R^2 + XRT)$, and for $\mu = 4$ from $|M|^4 \leq R^2 |M|^2$ applied with X^2 in place of X ; the B, C terms come from the mean value theorem $\int_0^T |Y^k|^2 \ll S^{o(1)}(T + Y^k)Y^k$. If Σ denotes the sum of the contributions, the route gives (1) on $\mathcal{T}_\sigma$ whenever $\alpha\tau + \Sigma \leq 1 + r - \delta'$ for a fixed $\delta' > 0$. When $\alpha > 0$ we write this as $\tau \leq (1 + r - \Sigma)/\alpha - \delta$, and when $\alpha = 0$ it reads $\Sigma \leq 1 + r - \delta$.

Definition 4.4. A point of the zeta domain is *zeta-certified* if some zeta route with $\alpha = 0$ satisfies $\Sigma \leq 1 + r - \delta$ there, or some zeta bound β and some zeta route with $\alpha > 0$ satisfy $\beta \leq (1 + r - \Sigma)/\alpha - \delta$ there, where $\delta = 10^{-6}$ as in Definition 4.2. A cell $[z', z''] \times [c', c'']$ is *zeta-admissible* if every point of the zeta domain with (z, c) in the cell is zeta-certified.

Proposition 4.5. Let $F = ZBC$, where Z is a zeta sum and B, C have no zeta factor and are each constant or of length at least z_1 , and suppose $(\log_x Z, \log_x C)$ lies, up to $o(1)$, in a zeta-admissible cell $[z', z''] \times [c', c'']$ with $z' > 1/4$. Then (1) holds.

Proof. If $Z \geq T_1 z_2$ and the coefficients of Z are ξ_0 or $\xi_0 \log$, this is [10, Lemma 5.4]. Otherwise (Z1) and [10, Lemma 5.8] place the point (z, c, v_Z, v_B, v_C) of each of the $S^{o(1)}$ sets $\mathcal{T}_\sigma$ in the zeta domain up to $o(1)$, the zeta bounds hold for τ up to the losses $x^{o(1)}$ and the loss $x^{\delta/2}$ of Lemma 2.2, and the zeta routes are valid up to $x^{o(1)}$. Since α takes finitely many values, a certified inequality $\beta \leq (1 + r - \Sigma)/\alpha - \delta$ gives $\alpha\beta + \Sigma \leq 1 + r - \alpha\delta$ with a fixed margin, and since all the forms involved are affine, the argument of Proposition 4.3 applies verbatim. $\square$

The zeta grid is computed by `coverZ.cpp` exactly as in the previous subsection, over $(z, c) \in [1/4, 1/2] \times [0, 1]$ with cells of side $1/400$. A zeta-case configuration is accepted if [10, Proposition 6.1(iii)] applies or its length box lies in a zeta-admissible cell; cells touching $z = 1/4$ are not used.

5. THE SIEVE

We use the procedure of [10, Section 7.2] with r in place of 0.07 (in particular with Lemma 3.1), and with the following additions. Throughout, a set of prime variables $p_1 > \dots > p_n$ in boxes $p_i \in [x^{\alpha_i}, x^{\alpha_i+h}]$ yields, by [10, Section 4], polynomials $F = P_1 \dots P_n \cdot W$ where W is the product of the remaining factors (R_i , the decomposed q -variables, and H), of length $x^{1-\sum \alpha_i + O(h)}$, whose

factors other than H are shorter than P_n , while H is either a zeta sum or shorter than $x^{1/4}$ ([10, Information 4.15, Remark 4.17]). By [10, Lemma 5.9] we may assume that F has at most one zeta factor.

- (a) *Zeta case.* If W contains a zeta factor, we test [10, Proposition 6.1(iii)] as in [10].
- (b) *Groupings.* If there is no zeta factor, we test every assignment of $P_1, \dots, P_n, W$ (and, in Järvinen's second strategy, of $P_1, \dots, P_n, R_1, W/R_1$ over a casework in $\log_x R_1$ with step $1/400$) to three nonempty groups A, B, C , and accept if the resulting length ranges satisfy [10, Proposition 6.1(i) or (ii)] or lie in admissible cells.
- (c) *Two variables.* For $n = 2$, [10] uses only the regions $B_1 \cup B_2$ of Lemma 3.1. We also apply tests (a), (b), (d) to the sum $\sum S(\mathcal{A}_{pq}, q)$; this is legitimate because the reduction of [10, Section 4] applies to any n .
- (d) *One Heath-Brown split.* By [10, Remark 4.10] the Heath-Brown decomposition may be applied to any subset of the prime variables. We apply it to a single P_i with $P_i \geq x^{1/4}$ and use Lemma 5.1.
- (e) *Splitting W .* In the case without a zeta factor every factor of W is shorter than x^m with $m = \max(\log_x P_n, 1/4)$ ([10, Information 4.15, Remark 4.17]). If $w = \log_x W > m$, then W has at least two factors, and the argument of Lemma 5.1 with $1/4$ replaced by m groups them as $W = W_1 W_2$ with $\log_x W_1 \in [w/2, \max(m, 2w/3)]$ and W_2 non-constant. On a box of prime variables the program evaluates m and w at the upper ends m_u, w_u of their ranges, and with w_l the lower end of the range of w it lets $\log_x W_1$ run over $[w_l/2, \max(m_u, 2w_u/3)]$, which contains $[w/2, \max(m, 2w/3)]$ at every point of the box. We run the grouping test (b) on $P_1, \dots, P_n, W_1, W_2$ with $\log_x W_1$ in steps of $1/400$ over this range, and accept if every step is accepted.
- (f) *Zeta factors.* Configurations with one zeta factor are also tested against the zeta grid of Section 4, and groupings containing a zeta factor are also tested against the main grid (Remark 4.1).

Lemma 5.1. *Let P have logarithmic length $p \in [1/4, 1/2)$ and apply the Heath-Brown decomposition. Each resulting product $P = N_1 \cdots N_J$ falls into one of two cases:*

- (i) *some N_j is a zeta sum Z , with $\log_x Z \in [1/4, p]$;*
- (ii) *every N_j is shorter than $x^{1/4}$, and the N_j can be grouped as $P = Q_1 Q_2$ with $\log_x Q_1 \in [p/2, \max(1/4, 2p/3)]$.*

Proof. A factor of length at least $x^{1/4}$ has coefficients $\xi_0, \xi_0 \log$ or $\xi_0 g$ by [10, Information 4.15], so it is a zeta sum, and there is at most one since $p < 1/2$. Otherwise let the lengths be $\ell_1 \geq \ell_2 \geq \dots$, all below $1/4$, with sum p . If $\ell_1 \geq 2p/3$, take $Q_1 = N_1$: then $\log_x Q_1 \in [2p/3, 1/4]$. If $\ell_1 \in [p/3, 2p/3)$, take Q_1 to be the longer of N_1 and P/N_1 . If $\ell_1 < p/3$, add factors in order until the sum first reaches $p/3$; the sum is then below $2p/3$, and we take Q_1 to be the longer of this product and its complement. In each case $\log_x Q_1 \in [p/2, \max(1/4, 2p/3)]$. $\square$

In case (i) we run a casework in $\log_x Z$ with step $1/400$ over $[1/4, p]$, and at each step test [10, Proposition 6.1(iii)] with $A = Z$ and the remaining factors P/Z , the other P_j , and W split into $B < x^{1/2}$, $C < x^{1/4+r}$ (if W also has a zeta factor we are done by [10, Lemma 5.9]). In case (ii) we run the grouping test (b) on $P_1, \dots, \hat{P}_i, \dots, P_n, Q_1, Q_2, W$, with $\log_x Q_1$ in steps of $1/400$ over the stated range; if W has a zeta factor, this configuration is tested by (a) and (f) instead. A box of prime variables is accepted if, for some i , every case is accepted.

The loss of a rejected box is bounded exactly as in [10, (7.6)–(7.7)]. The starting sum is over $x^r \leq q < p \leq x^{1/2-\epsilon}$ with boxes of side $1/3000$, and further Buchstab steps use side $1/400$, as in [10].

6. RESULTS OF THE COMPUTATION

The computation was run for $r = 0.061$, 0.059 and 0.0585 without steps (e), (f) and the extra routes, for $r = 0.058$ with step (e), and for $r = 0.0575$ with everything, using the 400×400 admissible grids. Table 1 lists the loss by range of $\log_x p$, where p is the larger of the two initial prime variables; entries are rounded up. With all steps, $r = 0.057$ fails: the ranges $[0.22, 0.5)$ already contribute 1.00221.

range of $\log_x p$	$r = 0.061$	$r = 0.059$	$r = 0.0585$	$r = 0.058$	$r = 0.0575$
$[r, 0.16)$	0.00025	0.00053	0.00064	0.00004	0.00004
$[0.16, 0.22)$	0.00887	0.01070	0.01155	0.01203	0.01070
$[0.22, 0.26)$	0.01000	0.01481	0.01698	0.01814	0.01691
$[0.26, 0.30)$	0.03605	0.04369	0.04679	0.04855	0.05011
$[0.30, 0.3337)$	0.11956	0.13434	0.13731	0.14142	0.13848
$[0.3337, 0.36)$	0.15231	0.15552	0.15830	0.15696	0.15760
$[0.36, 0.39)$	0.16113	0.16581	0.16711	0.16832	0.16808
$[0.39, 0.43)$	0.15523	0.16279	0.16594	0.16730	0.16914
$[0.43, 0.5)$	0.27063	0.28345	0.28663	0.28467	0.28053
total	0.91397	0.97161	0.99123	0.99738	0.99155

TABLE 1. Upper bounds for the sieve loss. Entries are rounded up; totals are rounded up from the unrounded sums.

Since the total loss is below 1, the argument of [10, Sections 3 and 7] gives, for all but $O(R)$ integers m , a lower bound $S(\mathcal{A}(m), 2\sqrt{x}) \gg \frac{\delta_0}{\delta_1} S(\mathcal{B}(m), 2\sqrt{x})$, and Theorem 1.1 follows from [10, Lemma 3.1] exactly as there.

Controls.

- (1) With $r = 0.07$, the tables and all additions switched off, our modified program reproduces Järvinen's total 0.99027 and his value 0.28464871 for the top range $[0.43, 0.5)$.
- (2) The admissible grids were spot-checked independently: for 3000 random admissible cells, 40 random points each (with (v_A, v_B, v_C) uniform in the domain (2)) were tested by a separate implementation; no failure occurred. A planted configuration $a = b = c = 1/3$, $\sigma = 0.63$, which is known not to be certified, is correctly rejected. For $r = 0.0575$, every certified cell (i, j) with $i + j < 399$ in either grid, where (i, j) denotes the cell with lower corner $(i/400, j/400)$, was tested (6186 cells in the main grid, and 15849 of the 15860 certified cells of the zeta grid, the other 11 lying on the outer diagonal $i + j = 399$), at 1500 points each by a separate implementation of the inequality lists, with no failure; on cells marked not admissible the same checker finds failing points in about 90% of samples.
- (3) With a grid built from Huxley's estimate alone (200×200 cells, direct treatment of two-variable sums, no full groupings and no Heath-Brown split) the loss at $r = 0.065$ exceeds 1.04, while the same procedure with Theorem 2.1 gives 0.922. So the Guth–Maynard estimate is what moves the exponent.
- (4) The complete computation for $r = 0.0575$, from both grids to the nine ranges of Table 1, was independently reproduced from the published code bundle, with total 0.99154810.

Code. The programs are available from the author on request and as ancillary files with the arXiv version, with a README that lists the command and environment flags behind each column of Table 1. The main grid is computed by `cover.cpp` (the plain grid, with the routes ρ_0 ,

ρ_X) and `coverM.cpp` (the extended Cauchy–Schwarz routes ρ_Q), which is run only on boundary cells, namely the cells not certified in the plain grid that lie within Chebyshev distance 2 of a certified cell. Every cell of the main grid for $r = 0.0575$ is certified in the plain grid or by `coverM.cpp` on a boundary cell with node budget 20000, and a script in the bundle rebuilds the cell-wise union of the two and checks this. The zeta grid is computed by `coverZ.cpp`. The loss is computed by `sieve3.cpp` (Järvinen's program with the changes of Section 5), `sieveE.cpp` (adding step (e)) and `sieveZ.cpp` (adding step (f)). The checkers are `verify_tab.py` and an independent pointwise checker for both grids at $r = 0.0575$, used in Controls (2).

7. LIMITS OF THE METHOD AND OF ZERO-BASED APPROACHES

This section records two limits on what can reach Andrica's conjecture itself. Theorem 7.6 shows that zero-freeness and $|t|^{o(1)}$ growth of the zeta function on $\sigma > 1/2$, together with error terms $O(x^{1/2+\varepsilon})$ for the integer and prime counting functions, are consistent with infinitely many failures; the construction says nothing about zeros on the critical line. Proposition 7.9 identifies a single-scale moment bound that would suffice.

7.1. Beurling systems. A Beurling system is a nondecreasing sequence $\mathcal{P} = (p_j)$ of reals $p_j > 1$ tending to infinity (the generalized primes); the generalized integers are the finite products, counted with multiplicity, with counting function $N(x)$; $\pi(x) = \#\{p_j \leq x\}$, $\psi(x) = \sum_{p_j^m \leq x} \log p_j$, and $\zeta_{\mathcal{P}}(s) = \prod_j (1 - p_j^{-s})^{-1} = \int_1^\infty u^{-s} dN(u)$ for $\sigma = \Re s > 1$; see [4] for the general theory.

Broucke, Debruyne and Vindas [2] constructed Beurling systems satisfying the Riemann hypothesis with large oscillation of $N(x)$. To control gaps between consecutive generalized primes we use the following theorem of Broucke and Vindas [3, Theorem 1.2].

Theorem 7.1 (Broucke–Vindas). *Let F be continuous, nondecreasing, with $F(1) = 0$, $F(x) \rightarrow \infty$ and $F(x) \ll x/\log x$. Then there is a strictly increasing sequence of generalized primes $\mathcal{P}$ with $|\pi_{\mathcal{P}}(x) - F(x)| \leq 1$ for all x , such that for all $x \geq 1$ and real t*

$$\left| \sum_{p_j \leq x} p_j^{-it} - \int_1^x u^{-it} dF(u) \right| \ll \sqrt{x} + \sqrt{\frac{x \log(|t| + 1)}{\log(x + 1)}},$$

where the implied constant depends on F and on $\mathcal{P}$.

In their construction, with $F(q_j) = j$, the prime p_j is drawn independently from dF restricted to $(q_{j-1}, q_j]$, and the bound holds almost surely. Below F satisfies $dF(W_k) = 0$ for a sequence of intervals W_k , so almost every realization also has no generalized prime in any W_k , and we choose such a realization.

7.2. The construction. Let L be nondecreasing with $1 \leq L(x) \leq \sqrt{x} \log x$ for all large x . Choose x_1 large and $x_{k+1} = x_k^2$. Put $W_k = (x_k, x_k + L_k]$ and $B_k = (x_k - L_k, x_k]$ with $L_k = L(x_k)$; these intervals are pairwise disjoint. Let

$$f(u) = \frac{1 - u^{-1}}{\log u} \left(1 + \sum_k \mathbf{1}_{B_k}(u) - \sum_k \mathbf{1}_{W_k}(u) \right) \quad (u > 1), \quad F(x) = \int_1^x f(u) du.$$

Then $f \geq 0$, $f = 0$ on each W_k , f doubles on B_k , F is continuous, nondecreasing and $F(x) \ll x/\log x$. Let $\mathcal{P}$ be a realization from Theorem 7.1 with no generalized prime in any W_k .

Lemma 7.2 (Gaps). *For each large k there are consecutive generalized primes $p < p'$ with*

$$x_k - 3 \log x_k < p \leq x_k, \quad x_k + L_k < p' \leq x_k + L_k + 3 \log x_k.$$

In particular $p' - p > L_k$.

Proof. No p_j lies in W_k , by the choice of realization. Since $f \geq (1 - u^{-1})/\log u$ on $I = (x_k - 3 \log x_k, x_k]$, which avoids every W_j , and $|\pi - F| \leq 1$, the interval I contains at least $F(x_k) - F(x_k - 3 \log x_k) - 2 \geq 3 - o(1) - 2 > 0$ generalized primes. The same argument on $I' = (x_k + L_k, x_k + L_k + 3 \log x_k]$, which also avoids every W_j , shows that I' contains a generalized prime. Let p be the largest generalized prime in I and p' the smallest in I' ; they are consecutive because none lies in W_k . $\square$

Lemma 7.3 (Primes). $\psi(x) = x + O(\sqrt{x} \log x)$.

Proof. Let $\psi_F(x) = \int_1^x \log u dF(u)$. Without the perturbation, $\int_1^x (1 - u^{-1}) du = x - 1 - \log x$. The k -th perturbation changes ψ_F by $\int_{B_k \cap [1, x]} (1 - u^{-1}) du - \int_{W_k \cap [1, x]} (1 - u^{-1}) du$, which has absolute value at most L_k and equals $\log(1 - L_k^2/x_k^2) = O(L_k^2/x_k^2) = O(\log^2 x_k/x_k)$ once $x \geq x_k + L_k$. Since $L_k \leq \sqrt{x_k} \log x_k$ and x_k grows doubly exponentially, $\psi_F(x) = x + O(\sqrt{x} \log x)$. By partial summation and $|\pi - F| \leq 1$, $\sum_{p_j \leq x} \log p_j = \psi_F(x) + O(\log x)$, and prime powers contribute $\ll \pi(\sqrt{x}) \log x \ll \sqrt{x}$. $\square$

Lemma 7.4 (Zeta function). *For every $\kappa \in (0, 1/2)$ and $\eta > 0$, $\zeta_{\mathcal{P}}(s)(s-1)$ extends analytically to $\sigma > 1/2$, $\zeta_{\mathcal{P}}(s) = A/(s-1) + O(1)$ near $s = 1$ with $A > 0$, and $|\zeta_{\mathcal{P}}(\sigma + it)| \ll_{\kappa, \eta} (|t| + 2)^\eta$ for $\sigma \geq 1/2 + \kappa$, $|t| \geq 1$.*

Proof. For $\sigma > 1$, $\log \zeta_{\mathcal{P}}(s) = \sum_j p_j^{-s} + Q(s)$ with $Q(s) = \sum_j \sum_{m \geq 2} p_j^{-ms}/m$, which is analytic and bounded on $\sigma \geq 1/2 + \kappa$ because $\pi(x) \ll x/\log x$. Next,

$$\sum_j p_j^{-s} = \int_1^\infty u^{-s} dF(u) + H(s), \quad H(s) = \int_1^\infty u^{-s} dE_t(u),$$

where $E_t(u) = \sum_{p_j \leq u} p_j^{-it} - \int_1^u v^{-it} dF(v)$. Integrating by parts and using Theorem 7.1, for $\sigma \geq 1/2 + \kappa$,

$$|H(s)| \leq \sigma \int_1^\infty |E_t(u)| u^{-\sigma-1} du \ll_\kappa 1 + \sqrt{\log(|t| + 2)}.$$

The function H is analytic on $\sigma > 1/2$: the truncations $H_Y(s) = \sum_{p_j \leq Y} p_j^{-s} - \int_1^Y u^{-s} dF(u)$ are entire, and by the same integration by parts $H_Y(s) = Y^{-s} E_t(Y) + \sigma \int_1^Y E_t(u) u^{-\sigma-1} du$, so Theorem 7.1 shows that $H_Y \rightarrow H$ locally uniformly on $\sigma > 1/2$ as $Y \rightarrow \infty$. Finally $\int_1^\infty u^{-s} dF(u) = \int_1^\infty u^{-s} \frac{1-u^{-1}}{\log u} du + G(s)$, where the first integral equals $\log \frac{s}{s-1}$ (differentiate in s), and

$$|G(s)| \leq \sum_k \int_{B_k \cup W_k} \frac{u^{-\sigma}}{\log u} du \ll \sum_k \frac{L_k x_k^{-1/2-\kappa}}{\log x_k} \ll \sum_k x_k^{-\kappa} < \infty,$$

uniformly in t , with G analytic on $\sigma > 1/2$. Hence $\zeta_{\mathcal{P}}(s) = \frac{s}{s-1} \exp(G(s) + H(s) + Q(s))$, which gives every claim with $A = \exp(G(1) + H(1) + Q(1))$, because $\exp(C\sqrt{\log(|t| + 2)}) \ll_\eta (|t| + 2)^\eta$. $\square$

Lemma 7.5 (Integers). $N(x) = Ax + O_\varepsilon(x^{1/2+\varepsilon})$ for every $\varepsilon > 0$.

Proof. Let $1 \leq y \leq x$ and $c_0 = 2$. By Perron's formula for the Mellin-Stieltjes transform,

$$\frac{1}{y} \int_x^{x+y} N(u) du = \frac{1}{2\pi i} \int_{c_0-i\infty}^{c_0+i\infty} \zeta_{\mathcal{P}}(s) \frac{(x+y)^{s+1} - x^{s+1}}{y s(s+1)} ds.$$

Move the line to $\sigma_0 = 1/2 + \kappa$, picking up the residue at $s = 1$, which is $A \frac{(x+y)^2 - x^2}{2y} = A(x + y/2)$. The horizontal segments vanish by Lemma 7.4. On $\sigma = \sigma_0$ the kernel is $\ll$

$x^{\sigma_0} \min(1/|s|, x/(y|s|^2))$, so the remaining integral is

$$\ll x^{\sigma_0} \left(\int_1^{x/y} t^{\eta-1} dt + \frac{x}{y} \int_{x/y}^{\infty} t^{\eta-2} dt + 1 \right) \ll x^{1/2+\kappa} (x/y)^{\eta}.$$

Since N is nondecreasing, $N(x) \leq \frac{1}{y} \int_x^{x+y} N \leq A(x+y/2) + O(x^{1/2+\kappa} (x/y)^{\eta})$, and the same argument on $[x-y, x]$ gives the lower bound. Take $y = x^{1/2+\kappa}$ and κ, η small. $\square$

Theorem 7.6. *For every nondecreasing L with $1 \leq L(x) \leq \sqrt{x} \log x$ for all large x there is a Beurling system with $N(x) = Ax + O_{\varepsilon}(x^{1/2+\varepsilon})$, $\psi(x) = x + O(\sqrt{x} \log x)$, $\zeta_{\mathcal{P}}(s)$ analytic and of growth $|t|^{o(1)}$ on $\sigma > 1/2$ apart from the simple pole at 1 (in particular nonvanishing there, being an exponential times $s/(s-1)$), and infinitely many pairs of consecutive generalized primes $p < p'$ with $p' - p > L(p)$.*

Proof. Lemmas 7.2–7.5; in Lemma 7.2, $p \leq x_k$ and L is nondecreasing, so $p' - p > L(x_k) \geq L(p)$. $\square$

Corollary 7.7. (i) *With $L(x) = 3\sqrt{x}$: a Beurling system satisfying the Riemann hypothesis for $\zeta_{\mathcal{P}}$, with $N(x) = Ax + O(x^{1/2+\varepsilon})$ and $\psi(x) = x + O(\sqrt{x} \log x)$, in which $\sqrt{p'} - \sqrt{p} \geq 1$ for infinitely many consecutive generalized primes. So Andrica's conjecture does not follow from these properties.*

(ii) *With $L(x) = \sqrt{x} \log x$: even $\psi(x) = x + O(\sqrt{x} \log x)$ together with $N(x) = Ax + O(x^{1/2+\varepsilon})$ allows gaps $p' - p > \sqrt{p} \log p$ infinitely often. So in this generality the size of the error term in ψ is also the size of the largest guaranteed-prime-free gap, up to a constant.*

Proof. (i) Take $p < p'$ as in Lemma 7.2, so that $p' - p > L_k = 3\sqrt{x_k}$, $p \leq x_k$ and $p' \leq x_k + 3\sqrt{x_k} + 3 \log x_k$, whence $\sqrt{p'} \leq \sqrt{x_k} + 2$. Then $\sqrt{p'} - \sqrt{p} = (p' - p)/(\sqrt{p'} + \sqrt{p}) > 3\sqrt{x_k}/(2\sqrt{x_k} + 2) > 1$. (ii) Immediate from Theorem 7.6. $\square$

Remark 7.8. Corollary 7.7(i) shows that any argument using only the properties listed in Theorem 7.6, namely zero-freeness and $|t|^{o(1)}$ growth of the zeta function on $\sigma > 1/2$ and the stated asymptotics for N and ψ , cannot prove Andrica's conjecture. The construction does not model the zeros on the critical line, so arguments that use finer information about them, such as their pair correlation, are not covered. Any proof must use input of that kind or arithmetic of $\mathbb{Z}$ that Beurling systems lack, such as the additive structure of the integers or bilinear information about integers in short intervals. Sieve methods do use such information, but their Type II input collapses at length $\sqrt{x}$.

7.3. A moment hypothesis that implies Andrica.

Proposition 7.9. *Let $k > 2$ be fixed and suppose that for all large X , with $H = \sqrt{X}$,*

$$\int_X^{2X} |\psi(y+H) - \psi(y) - H|^k dy < 2^{-k} X^{(k+1)/2}.$$

Then $\sqrt{p_{n+1}} - \sqrt{p_n} < 1$ for all sufficiently large n .

Proof. Suppose $(x, x + 2\sqrt{x} + 1]$ contains no prime, and choose X with $x \in [X, 1.5X]$, so that $H = \sqrt{X} \leq \sqrt{x}$. For $y \in [x, x+H]$ the interval $(y, y+H]$ lies in $(x, x + 2\sqrt{x}]$, so it contains no prime and at most $O(\log x)$ prime powers ($O(1)$ for each exponent $m \leq \log_2(2x)$), contributing $O(\log^2 x)$ to ψ . Hence $|\psi(y+H) - \psi(y) - H| \geq H/2$ on a set of measure H inside $[X, 2X]$, and the integral is at least $H(H/2)^k = 2^{-k} X^{(k+1)/2}$, a contradiction. $\square$

Remark 7.10. The method is folklore for $k = 2$, where it fails: the conjectured variance is $\asymp X^{3/2} \log X$, a factor $\log X$ too large. For $k = 4$, Conjecture 1 of Montgomery and Soundararajan

[11] states that $\sum_{n \leq N} (\psi(n+H) - \psi(n) - H)^4 \sim 3NH^2 \log^2(N/H)$ uniformly for $(\log N)^{1+\kappa} \leq H \leq N^{1-\kappa}$, for each fixed $\kappa > 0$. Taking the difference of the cases $N = 2X$ and $N = X$ with $H = \sqrt{X}$ gives $\sum_{X < n \leq 2X} (\psi(n+H) - \psi(n) - H)^4 \sim \frac{3}{4}X^2 \log^2 X$. For $y \in [n, n+1)$ the quantity $\psi(y+H) - \psi(y) - H$ differs from its value at n by $O(\log X)$, and by Minkowski's inequality this changes the fourth root of the fourth moment by $O(X^{1/4} \log X)$, which is negligible. Hence the conjecture predicts $\int_X^{2X} |\psi(y+H) - \psi(y) - H|^4 dy \sim \frac{3}{4}X^2 \log^2 X$, well below $X^{5/2}/16$. So that conjecture implies Andrica's conjecture for large n ; together with the verification below 2^{64} [12], it would settle it, provided the threshold is made explicit. No unconditional bound of the required strength is known, and by Theorem 7.6 none can follow from the properties listed there alone.

8. CONCLUDING REMARKS

The largest remaining contribution, about 0.28, comes from $p \in [x^{0.43}, x^{1/2}]$, almost entirely from boxes without a zeta factor in which no grouping lands in the admissible set. Stronger large values estimates near $V \approx N^{3/4}$, or an improvement of Heath-Brown's mean value theorem, would directly lower the exponent. Andrica's conjecture and Legendre's conjecture remain open; Section 7 records what any proof must supply.

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